

# Acquisition and Analysis of Neural Data

## Problem Set 2.

Shokrofeh J Arina

AI has been used to review the solutions.

---

1/ Convolution and Cross-Correlation / Time duration: 40 min

$$(x * y)(t) = \int_{-\infty}^{+\infty} ds \, x(s) y(t-s). \quad \leftarrow \text{Convolution}$$

$$(x \star y)(t) = \int_{-\infty}^{+\infty} ds \, x(s) y(t+s) \quad \leftarrow \text{Cross-Correlation}$$

a) how are convolution and cross-correlation related to each other?

Convolution flips  $(-s)$  the kernel before sliding it across the data.

In fact, Convolution is Correlation with the filter rotated by  $180^\circ$ .

Write the cross-correlation  $(x \star y)(t)$  as a convolution  $(\tilde{x} * y)(t)$

$$(x \star y)(t) = \int_{-\infty}^{\infty} d(s) x(s) y(t+s) \quad u = -s \Rightarrow s = -u \\ ds = -du$$

$$\text{Substitute: } (x \star y)(t) = \int_{+\infty}^{-\infty} (-du) x(-u) y(t-u) \quad \begin{cases} s \rightarrow -\infty, u = +\infty \\ s \rightarrow +\infty, u = -\infty \end{cases}$$

$$\text{flipping the limit: } = \int_{-\infty}^{+\infty} (du) x(-u) y(t-u)$$

$$\Rightarrow \tilde{x}(u) = x(-u) \Rightarrow (x \star y)(t) = \int_{-\infty}^{+\infty} du \tilde{x}(u) y(t-u) = (\tilde{x} * y)(t)$$



- How is the new function  $\tilde{x}$  related to the original function  $x$ ?
- When are the two operations equivalent?

$\tilde{x}$  is a time-reversed version of  $x$ .

They are both equivalent when reversing  $x$  does not change it.

$x(t) = x(-t) \rightarrow$  When  $x$  is symmetric like  
a Gaussian centered at zero  
is a symmetric and even.

Cross-correlation is like convolution, but before convolving, one  
of signal is flipped in time.

$$b) x(t) := \left[ \frac{t}{t_{\text{spike}}} e^{-t/t_{\text{spike}}} \right]_+ \quad \text{alpha function}$$

$$y(t) := \delta(t - t_s) \quad \text{Dirac function}$$

Compute:  $(x * y)(t)$  and  $(x \star y)(t)$ . sketch the result.

Given:

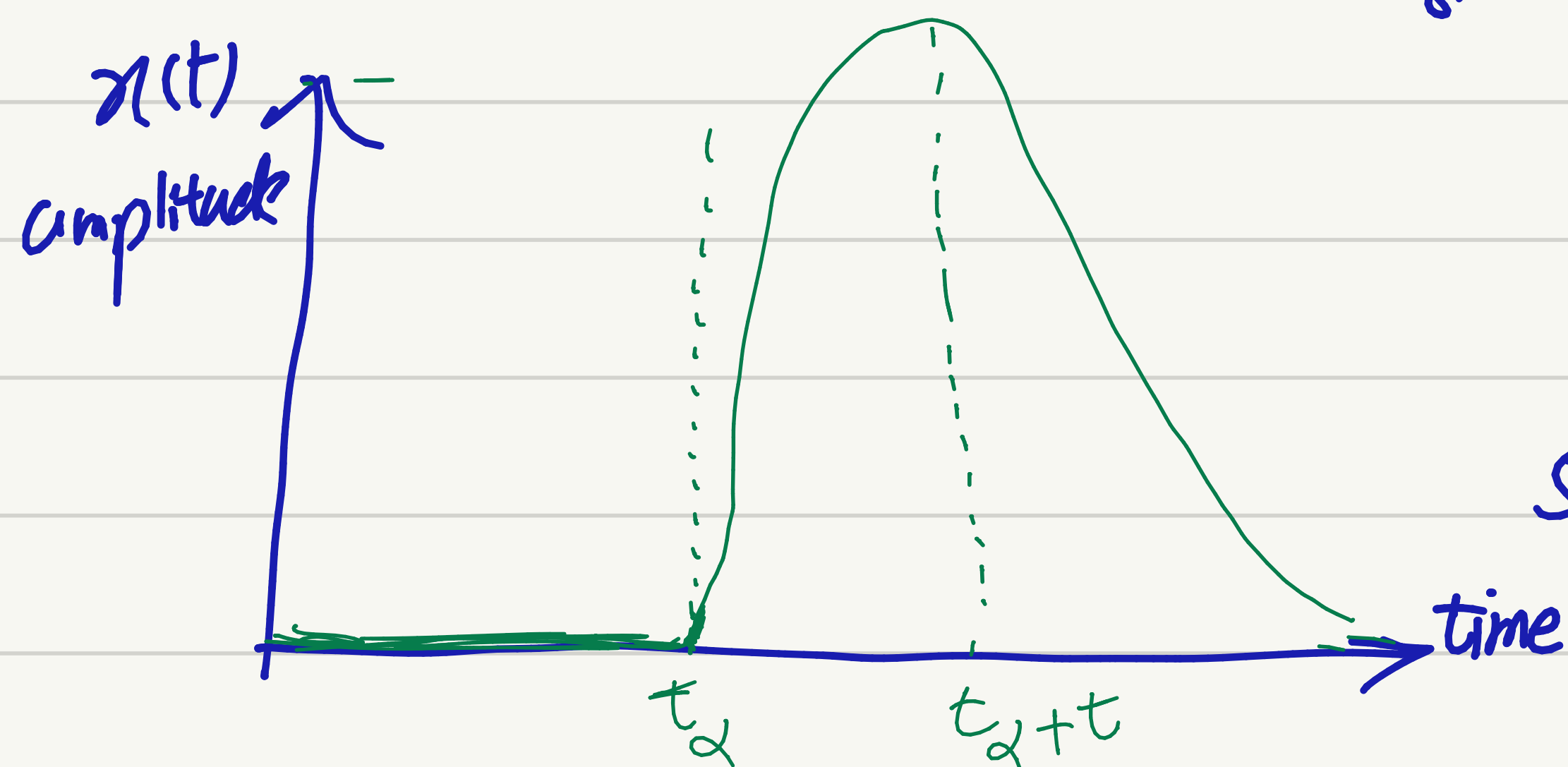
$$(x * y)(t) = \int_{-\infty}^{+\infty} ds x(s) y(t-s)$$

$$y(t) = \delta(t - t_s)$$

$$y(t-s) = \delta((t-s) - t_s)$$

$$t-s-t_s = -(s-(t-t_s))$$

$$\delta(-z) = \delta(z) \Rightarrow \delta(t-s-t_s) = \delta(s-(t-t_s))$$



$$(x * y)(t) = \int_{-\infty}^{\infty} ds x(s) \delta(s - \underbrace{(t-t_s)}_a)$$

$$\Rightarrow \text{delta property: } \int_{-\infty}^{\infty} ds f(s) \delta(t-a) = f(a)$$

$$\Downarrow$$

$$(x * y)(t) = x(t-t_s)$$

$$(x * y)(t) = \begin{cases} \frac{t-t_s}{t_{\text{spike}}} e^{-\frac{t-t_s}{t_{\text{spike}}}} & t \geq t_s \\ 0 & t < t_s \end{cases}$$

so the Convolution with  $\delta(t-t_s)$  shifts the alpha function to the right by  $t_s$



$$(x \star y)(t) = \int_{-\infty}^{\infty} ds x(s) y(t+s)$$

$$y(t) = \delta(t - t_\delta)$$

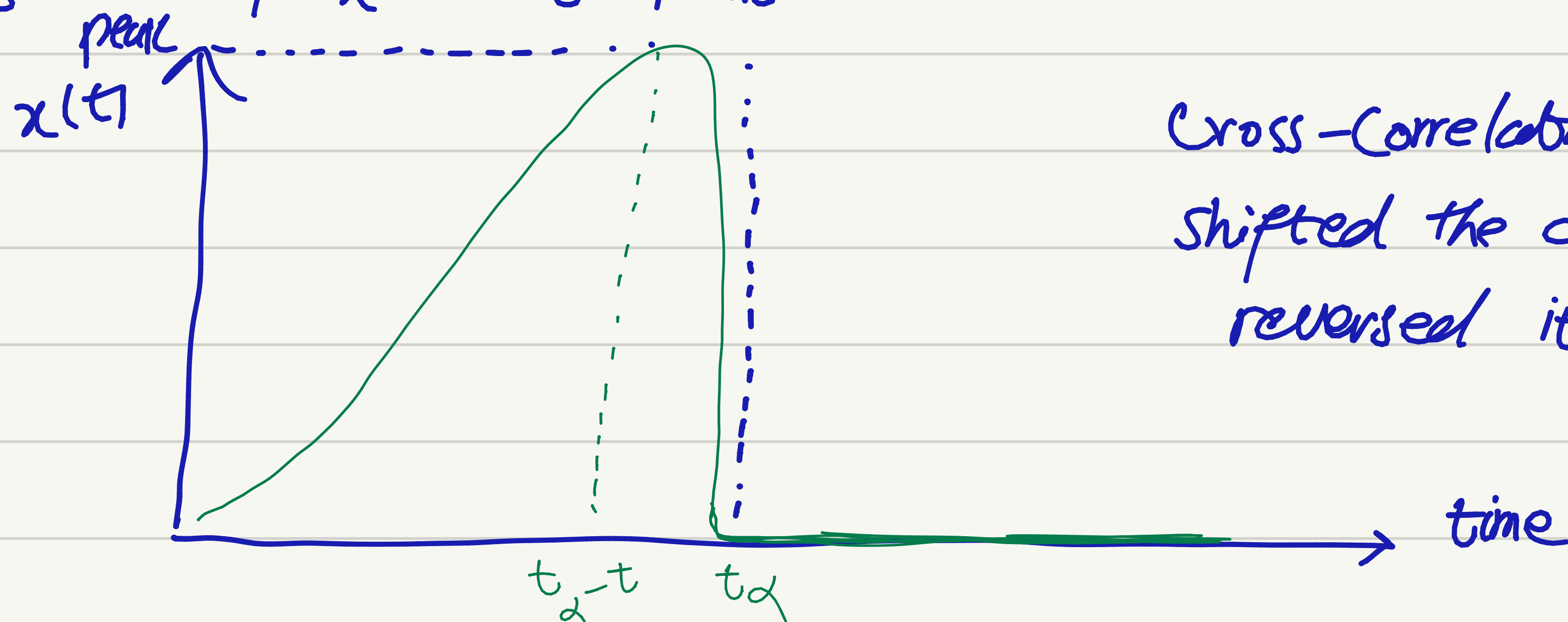
$$y(t+s) = \delta((t+s) - t_\delta)$$

$$t+s-t_\delta = s - (t_\delta - t)$$

$$\delta(t+s-t_\delta) = \delta(s - (t_\delta - t))$$

$$t_\delta - t \geq 0 \Rightarrow t \leq t_\delta$$

$$t_\delta - t = t_{\text{peak}} \Rightarrow t_\delta + t_{\text{peak}} = t$$



$$(x \star y)(t) = \int_{-\infty}^{\infty} ds x(s) \delta(s - \underbrace{(t_\delta - t)}_a)$$

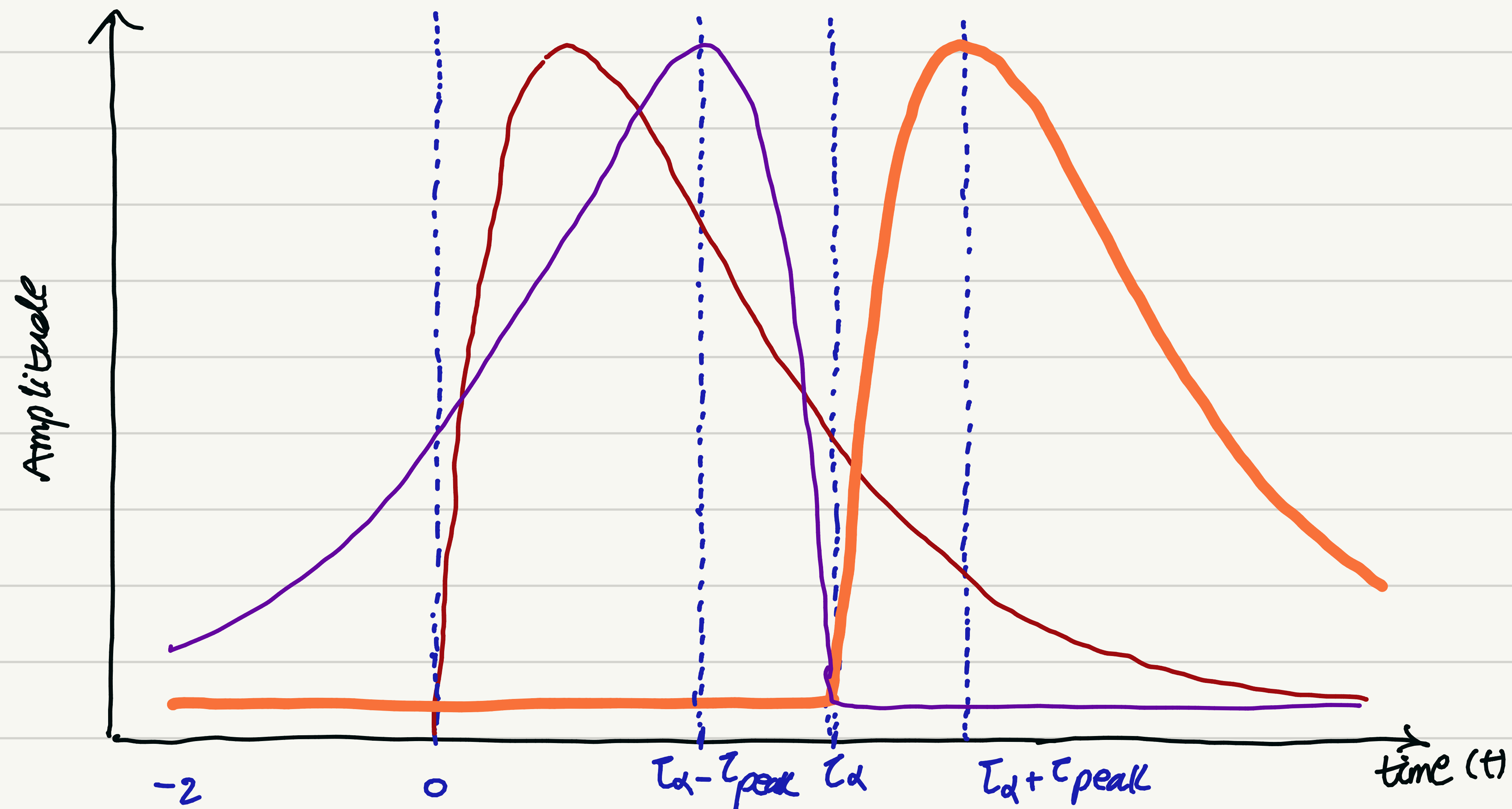
$$\Rightarrow \text{delta fun property: } \int_{-\infty}^{\infty} ds f(s) \delta(s-a) = f(a)$$

$$(x \star y)(t) = x(t_\delta - t)$$

$$(x \star y)(t) = \begin{cases} \frac{t_\delta - t}{t_{\text{spike}}} e^{-\frac{t_\delta - t}{t_{\text{spike}}}} & t \leq t_\delta \\ 0 & t > t_\delta \end{cases}$$

Cross-correlation with  $\delta(t - t_\delta)$  shifted the alpha function and reversed it.

- original  $x(t)$
- Convolution  $(x * y)(t) = x(t - t_g)$
- Cross-Correlation  $(x \star y)(t) = x(t_g - t)$



Convolution and Cross-Correlation with a Dirac delta



c) Suppose you find a pronounced peak in the cross-correlation between two spike trains. How would you interpret this findings? Time: 20 min

The cross-correlation between two spike trains means that the two neurons tend to fire with a consistent temporal relationship.

$C_{xy}(\tau)$ , measure how often spikes in neuron  $y$  occur at a time lag  $\tau$  relative to spikes in neuron  $x$ .

if the peak is at  $\tau=0 \Rightarrow$  two neuron fire synchronously

if the peak is at  $\tau > 0 \Rightarrow$  neuron  $y$  tends to fire after  $x$  with approximation of  $\tau$

if the peak is at  $\tau < 0 \Rightarrow$  neuron  $y$  tends to fire before  $x$  with approximation of  $\tau$

Cross-correlation is a measure of statistical dependency, not causality. You cannot distinguish these scenarios from cross-correlation alone. But comparing the peak lag  $\tau$  to known synaptic delay can give a hint.

A typical monosynaptic excitatory connection has a delay of  $1 \sim 3$  ms.



axonal conduction + synaptic transmission. That means if you see a peak at  $\tau \approx 1-3$  ms, This is consistent with a direct synaptic connection.

$x \rightarrow y$ .

if  $\tau$  is much larger (e.g. 10-20 ms) it suggests a poly synaptic or common-input scenario (e.g. a third neuron  $z$  drives both, but with slightly different delays.

if  $\tau = 0$  with a very sharp peak, common input (synchronizing both neurons simultaneously) is more likely than a direct connection.



## 2) Normalization of probability density

Time duration: 5 min

$$\int_{-\infty}^{\infty} dx p(x) = 1$$

$$x(t) = \begin{cases} At e^{-at} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$\int_0^{\infty} At e^{-at} dt = 1 = A \int_0^{\infty} t e^{-at} dt = 1 \quad \star \quad \int u dv = uv - \int v du$$

$$\begin{cases} u = t \\ du = dt \end{cases} \quad \begin{cases} e^{-at} dt = dv \\ -\frac{1}{a} e^{-at} = v \end{cases}$$

$$\int_0^t t e^{-at} dt = t \left(-\frac{1}{a}\right) e^{-at} - \int -\frac{1}{a} e^{-at} dt = -\frac{t}{a} e^{-at} + \frac{1}{a} \int_0^{\infty} e^{-at} dt$$

$$= -\frac{t}{a} e^{-at} + \frac{1}{a^2} e^{-at} \Big|_0^{\infty}$$

$$= \frac{-\infty}{a} e^{-\infty} - \frac{1}{a^2} e^{-\infty} - \left( \frac{-0}{a} e^{-0} - \frac{1}{a^2} e^{-0} \right)$$

$$= \frac{1}{a^2}$$

substitute to  $\star$   $A \left( \frac{1}{a^2} \right) = 1 \Rightarrow A = a^2$ ,  $x(t) = \left[ a^2 t e^{-at} \right]_+$

### 3/ Coefficient of variation of a poisson process.

a) show that  $p_{ISI}(\tau)$  is a normalized probability density time: 5min

$$\int_{-\infty}^{+\infty} p(x) dx = 1$$

$$p_{ISI}(\tau) = \begin{cases} re^{-r\tau} & \tau > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\int_0^{\infty} re^{-r\tau} d\tau = 1$$

$$r \int_0^{\infty} e^{-r\tau} d\tau = 1$$

$$r \left( -\frac{1}{r} e^{-r\tau} \right) \Big|_0^{\infty} = 1 \Rightarrow r \left( \lim_{\tau \rightarrow \infty} \frac{1}{r} e^{-r\tau} - \left( \frac{1}{r} e^0 \right) \right) = r \left( \frac{1}{r} \right) = 1$$

Therefore:  $\int_0^{\infty} re^{-r\tau} d\tau = 1$  is a normalized probability density



b) Compute the mean  $\langle \tau \rangle$ , and the variance  $\sigma_\tau^2$  of the inter-events intervals  $\tau$ .  
 Time: 70 min

first moment  $\langle x \rangle := \int_{-\infty}^{\infty} dx x p(x)$

$$\langle \tau \rangle := \int_0^{\infty} \tau r e^{-r\tau} d\tau = r \int_0^{\infty} \tau e^{-r\tau} d\tau$$

$$\int u dv = uv - \int v du$$

$$= r \left[ \frac{-\tau}{r} e^{-r\tau} - \int \frac{-1}{r} e^{-r\tau} d\tau \right] \Big|_0^{\infty} \quad \left\{ \begin{array}{l} u = \tau \\ du = d\tau \end{array} \right. \quad \left\{ \begin{array}{l} e^{-r\tau} d\tau = dv \\ -\frac{1}{r} e^{-r\tau} = v \end{array} \right.$$

$$= r \left[ -\frac{\tau}{r} e^{-r\tau} \Big|_0^{\infty} + \frac{1}{r^2} e^{-r\tau} \Big|_0^{\infty} \right]$$

$$= r \left[ \cancel{\frac{-\tau}{r} e^{-r\tau} \Big|_0^{\infty}} + \frac{1}{r^2} e^{-r\tau} \Big|_0^{\infty} \right]$$

$$e^{-\infty} \rightarrow 0$$

$$e^0 \rightarrow 1$$

$$r \left[ 0 + \frac{1}{r^2} (0 - 1) \right] = r \cdot \frac{1}{r^2} = \frac{1}{r}$$

Therefore:  $\langle \tau \rangle = \frac{1}{r}$

$\langle \tau \rangle = \frac{1}{r}$  means that if the firing rate is high the average waiting time between spike is short.



The second moment  $\langle x^2 \rangle := \int dx x^2 p(x)$

Time: 70 min

$$\langle \tau^2 \rangle := \int_0^\infty \tau^2 r e^{-r\tau} d\tau$$

$$\int u dv = uv - \int v du$$

$$= r \left[ \frac{-\tau^2}{r} e^{-r\tau} - \int_0^\infty \frac{-1}{r} e^{-r\tau} (2\tau) d\tau \right]$$

$$\begin{cases} \tau^2 = u \\ 2\tau d\tau = du \end{cases} \begin{cases} e^{-r\tau} d\tau = dv \\ \frac{-1}{r} e^{-r\tau} = v \end{cases}$$

$$= r \left[ \cancel{\frac{-\tau^2}{r} e^{-r\tau}} \Big|_0^\infty + \frac{2}{r} \int_0^\infty \tau e^{-r\tau} d\tau \right]$$

$\frac{1}{r^2}$  from previous part for  $\langle x \rangle$

$$= r \left[ \frac{2}{r} \cdot \frac{1}{r^2} \right] = \frac{2}{r^2}$$

$$\text{Therefore } \langle \tau^2 \rangle = \frac{2}{r^2}$$

$$\sigma_\tau^2 := \langle \tau^2 \rangle - \langle \tau \rangle^2 = \left( \frac{2}{r^2} \right) - \left( \frac{1}{r^2} \right) = \frac{1}{r^2}, \quad \sigma_\tau = \frac{1}{r}$$

$\sigma_\tau$  means how much the inter-spike intervals fluctuate around the mean  $\langle \tau \rangle$ , and  $\sigma_\tau = \langle \tau \rangle$ .  
In Poisson process the mean and the size of fluctuation is the same, meaning the spike train is irregular.  
Some intervals may be much shorter and some may be much longer.

c) Compute the Coefficient of Variation CV of the Poisson process.

$$\sigma_{\tau} = \frac{1}{r}, \quad \langle \tau \rangle = \frac{1}{r}$$

$$CV = \frac{\sigma_{\tau}}{\langle \tau \rangle}, \quad \text{Time: 3 min}$$

$$CV = \frac{\sigma_{\tau}}{\langle \tau \rangle} = \frac{\frac{1}{r}}{\frac{1}{r}} = 1$$

CV is independent of firing rate and standard deviation is equal

to the mean  $\langle \tau \rangle$ .  $\sigma_{\tau} = \langle \tau \rangle$

$CV = 1 \Rightarrow$  poisson like, random firing.

$CV > 1 \Rightarrow$  more irregular or bursty firing than poisson.

$CV < 1 \Rightarrow$  more regular firing than poisson.



#### 4/ Moments of the Poisson Distribution 8

Time : 30 min

$$\langle n \rangle = \sum_{n=0}^{\infty} n P_T[n] = \sum_{n=0}^{\infty} \frac{n (rT)^n}{n!} \exp(-rT)$$

$$\sigma_n^2 \langle T \rangle = \sum_{n=0}^{\infty} n^2 P_T[n] - \langle n \rangle^2 = \sum_{n=0}^{\infty} \frac{n^2 (rT)^n}{n!} \exp(-rT) - \langle n \rangle^2$$

To calculate mean and Variance is to to compute the moment-generating function.

$$g(a) = \sum_{n=0}^{\infty} \frac{(rT)^n \exp(an)}{n!} \exp(-rT)$$

$$\left. \frac{d^k g}{da^k} \right|_{a=0} = \sum_{n=0}^{\infty} \frac{n^k (rT)^n}{n!} \exp(-rT)$$

and once we computed  $g$ , we need to calculate only first and second derivative to determine the sum we need.

So from the previous part (Normalization) we have:  $\int_0^{\infty} e^{-r\tau} d\tau = \frac{1}{r}$

Then differentiate both side with respect to  $r$ :

$$\frac{d}{dr} \int_0^{\infty} e^{-r\tau} d\tau = \frac{d}{dr} \left( \frac{1}{r} \right)$$

$$\int_0^{\infty} -\tau e^{-r\tau} d\tau = -\frac{1}{r^2} \Rightarrow \int_0^{\infty} \tau e^{-r\tau} d\tau = \frac{1}{r^2}$$

for the second Moment again differentiate

$$\frac{d}{dr} \left( \int_0^{\infty} \tau e^{-r\tau} d\tau \right) = \frac{d}{dr} \left( \frac{1}{r^2} \right) =$$

$$\int_0^{\infty} -\tau^2 e^{-r\tau} d\tau = -\frac{2}{r^3} \Rightarrow \int_0^{\infty} \tau^2 e^{-r\tau} d\tau = \frac{2}{r^3}$$

$$\text{Then : } \langle \tau \rangle = r \int_0^{\infty} \tau e^{-r\tau} d\tau = r \cdot \frac{1}{r^2} = \frac{1}{r} \quad \langle \tau^2 \rangle = r \int_0^{\infty} \tau^2 e^{-r\tau} d\tau = r \cdot \frac{2}{r^3} = \frac{2}{r^2}$$